

L04 CONTRIBUTION JOURNAL

ALEXANDER DEBUSK

ITAI 1371 — Module 04: Exploratory Data Analysis

Lab 4 – Contributions

Coordinated tight effort to have everyones files submitted on time for both Lab 3 and Lab 4

Finished my own work on time as well.

L04 CONTRIBUTION JOURNAL
SHAREEFAH ADEOLA

ITAI 1371 — Module 04: Exploratory Data Analysis

CONTRIBUTION JOURNAL

ITAI 1371 – Module 04: Exploratory Data Analysis

Group: ML_1371_14274 4

Lab: Module 04 – Exploratory Data Analysis

Project: Titanic Dataset – Exploratory Data Analysis

Student: Shareefah Adeola

Purpose of the Contribution Journal

The project involved examining the Titanic dataset, performing descriptive statistical analysis, creating visualizations, conducting additional experiments, interpreting the results, completing the Knowledge Check, and preparing the required documentation for submission.

I completed my portion of the Module 04 project independently and worked through the notebook systematically in Google Colab.

Documentation and Submission Preparation

I also:

- Reviewed the completed notebook.
- Checked that the required sections were completed.
- Reviewed the outputs and visualizations.
- Prepared the work for PDF submission.
- Prepared the required reflective and contribution journals.

My contribution therefore covered the **execution, exploration, visualization, interpretation, experimentation, Knowledge Check, and documentation** portions of the L04 project. The completed notebook confirms these activities and the associated analysis.

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YILIN LENG

ITAI 1371 — Module 04: Exploratory Data Analysis

Contribution

For this Module 04 Lab, I followed the notebook to complete the tasks. I ran the required code and checked the outputs.

I mainly worked on the "Student Experimentation" section. In Experiment 1, I used a countplot to look at the relationship between the Port of Embarkation and survival. I compared the results for the three locations: C, Q, and S. After creating the chart, I wrote my own insight based on what I saw in the visualization.

In Experiment 2, I used a boxplot to compare the fare distributions of survivors and non-smokers. I noticed that survivors generally paid higher fares. I also looked at how this result may be related to passenger class and survival.

I also answered the three "Knowledge Check" questions and checked the charts and outputs in the notebook to make sure the code worked correctly and the results were shown clearly.

During this lab, I did not only check if the code worked. I also took time to understand what each visualization showed. My main contributions were completing the two experiments, analyzing the results, answering the Knowledge Check questions, and checking the final notebook content.

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KEITH JONES

ITAI 1371 — Module 04: Exploratory Data Analysis

Contribution Journal — Titanic EDA Lab

A record of the work I completed and the decisions I made at each stage of this assignment.

Task: Countplot of Survival by Port of Embarkation

Wrote the `sns.countplot()` call comparing Embarked against Survived.

Added axis labels and a legend clarifying the port codes (C/Q/S) so the plot is readable without needing to reference the assignment text.

Went beyond the countplot by computing `df.groupby('Embarked')['Survived'].mean()` to get an actual survival rate per port rather than relying on eyeballing raw counts, since Southampton had far more passengers than the other two ports.

Interpreted the result: Cherbourg passengers showed a higher survival rate, and flagged passenger class as a likely confound worth checking rather than treating port as a direct cause.

Task: FacetGrid of Age Distribution by Survival

Debugged a broken code cell where an uncommented explanatory sentence was going to raise a `SyntaxError`; rewrote it as proper multi-line comments so the cell would run.

Ran the `FacetGrid/plt.hist` comparison of Age across survivors and non-survivors.

Identified two things the raw plot didn't account for: missing Age values being silently dropped, and the two panels having different total sample sizes (more passengers died than survived), which affects how the shapes should be compared.

Proposed a quantitative follow-up (`df['is_child'] = df['Age'] < 12` and a `groupby`) to turn the visual "spike for children" observation into an actual number.

Task: Fare vs. Survival (Boxplot/Violinplot)

Built both a boxplot and a violinplot of Fare by Survived, since the instructions offered a choice and I wanted to compare what each revealed.

Diagnosed that the raw Fare distribution's heavy right skew was squashing both plots near zero, and fixed it two ways: capping the y-axis and, as an alternative, log-scaling it.

Computed median (not mean) fare by survival group, explicitly choosing the median because the skewed distribution would let a handful of expensive tickets distort the mean.

Connected this result back to the earlier port finding, noting that fare and port both likely reflect class rather than being independent effects.

Task: Reflection Questions (Markdown Cell)

Answered the EDA-purpose question by framing EDA as exploratory pattern-finding and quality-checking prior to modeling, not confirmation of a hypothesis.

Synthesized the three prior experiments into a single answer to "what kind of passenger had the best survival odds" — young, female, first class — rather than restating each plot's result separately.

Answered the visualization-vs-statistics question with a concrete argument about what summary numbers hide (skew, rate vs. count, confounds), and referenced Anscombe's Quartet as a citable example of the

general principle.

Task: Reflection and Journal Write-Up

Wrote a reflection identifying the pattern that ran across all four experiments: class as a likely common cause underlying the port, fare, and survival relationships, and how that only became visible by comparing plots rather than reading any one of them in isolation.

Wrote a chronological journal documenting the process for each experiment, including bugs encountered and fixed along the way, not just final results.

Overall contribution summary: I completed all coding, debugging, statistical follow-up, and written analysis for this lab — including catching and fixing two code errors that would have prevented cells from running (the tuple-indexing bug in the port experiment and the unclosed comment in the FacetGrid cell), and extending each visualization with a numeric check (survival rate, median fare, child-survival rate) rather than relying on visual inspection alone.